Bounded constraint-graph transducer certificates for arithmetic capacity

RH research program

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Abstract

We isolate a finite, reusable theorem behind several arithmetic capacity experiments. A finite directed constraint graph with a bounded vertex observable admits an exact open-prefix max-plus dynamic program. A clocked finite-memory transducer is called universally safe when every possible pair of inputs produces a legal graph edge. If its observed label is a one-site factor of the current clock phase and the current Mobius value, then its Mobius correlation has an unconditional arithmetic-progression formula in terms of squarefree densities. Consequently every such certificate gives a rigorous lower bound on the graph capacity liminf, and all certificates with fixed clock and memory budgets form a finite enumerable class. We verify the contract on the RH-366 four-state graph, the distinct RH-368 three-cell factor, and a new period-three safe switch on the RH-366 graph. Labels that depend on memory are explicitly left open because they require higher-order Mobius correlations. This is a bounded classification result, not a classification of all subshifts and not a spectral or zeta construction.

1 Frozen data and claim boundary

RH-366 fixes a four-state survivor whose state order is $(--, -+, +-, ++)$ and whose adjacency matrix is

$$A_{366} = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \qquad \ell_{366} = (-1, -1, 1, 1). \tag{1}$$

The graph path records a pair of consecutive signs, so this is the distance-two constraint used by RH-366 [1]. RH-368 supplies a different three-cell graph

$$A_{368} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \qquad \ell_{368} = (1, -1, -1), \tag{2}$$

with its parity factor and all-order capacity theorem [2]. We use both only as frozen finite graph inputs; the two languages are not identified.

For a finite directed graph $G = (V, E)$ and an integer observable $\ell : V \to \mathbb{Z}$, write

$$K_N(G, \ell) = \max_{\substack{v_1,\dots,v_N \in V \\ (v_n, v_{n+1}) \in E}} \left|\sum_{n=1}^{N} \mu(n)\ell(v_n)\right|. \tag{3}$$

The optimizing open path may depend on the complete prefix. This definition is a finite scalar functional; it is not a trace or a determinant.

2 Exact graph capacity

Theorem 2.1 (Open-path max-plus recurrence)**.** *Assume every vertex has an incoming edge (as in the frozen graphs above). Put*

$$D_1^{\pm}(v) = \pm\mu(1)\ell(v), \qquad D_{n+1}^{\pm}(v) = \pm\mu(n+1)\ell(v) + \max_{(u,v)\in E} D_n^{\pm}(u). \tag{4}$$

Then

$$K_N(G,\ell) = \max_{v\in V} \max\{D_N^+(v), D_N^-(v)\}. \tag{5}$$

The recurrence and a maximizing witness require $O(N|E|)$ integer operations.

Proof. Induct on n. The value $D_n^+(v)$ is the largest signed score among paths ending at v: appending v contributes $\mu(n)\ell(v)$, and the predecessor must be an incoming neighbor. The same induction with the opposite sign gives $D_n^-(v)$, the largest negative of a path score. Taking the maximum over terminal vertices proves (5). Each edge is inspected once per step; predecessor pointers give a witness. □

Corollary 2.2 (Universal squarefree upper bound)**.** *For bounded ℓ,*

$$\limsup_{N\to\infty} \frac{K_N(G,\ell)}{N} \le \frac{6}{\pi^2}\|\ell\|_\infty. \tag{6}$$

Proof. Every admissible path has absolute score at most $\|\ell\|_\infty \sum_{n\le N} \mu(n)^2$. The squarefree density is $6/\pi^2$. □

3 Safe finite-memory transducers

Definition 3.1 (Clocked transducer)**.** Fix a clock $q \ge 1$, a finite state set S, and input alphabet $\mathcal{A} = \{-1, 0, 1\}$. A transducer consists of maps

$$\tau : S \times \mathbb{Z}/q\mathbb{Z} \times \mathcal{A} \to S, \qquad \omega : S \times \mathbb{Z}/q\mathbb{Z} \times \mathcal{A} \to V.$$

At time n, with $a_n = \mu(n)$ and $r_n = n \bmod q$, it outputs $v_n = \omega(s_n, r_n, a_n)$ and updates $s_{n+1} = \tau(s_n, r_n, a_n)$.

Definition 3.2 (Universal safety and one-site factor)**.** The table (τ, ω) is universally safe on G if, for every $s \in S$, $r \in \mathbb{Z}/q\mathbb{Z}$, and $a, b \in \mathcal{A}$,

$$(\omega(s,r,a), \omega(\tau(s,r,a), r+1, b)) \in E. \tag{7}$$

It has a one-site observed factor if there are functions $g_r : \mathcal{A} \to \mathbb{Z}$ with

$$\ell(\omega(s,r,a)) = g_r(a) \quad \text{for every } s, r, a. \tag{8}$$

The universal quantifier in (7) is intentional. It is a finite table check and does not assume anything about future Mobius values. The one-site condition is the arithmetic firewall: the internal memory may choose a legal path, but it may not alter the observed label.

Lemma 3.3 (Squarefree density in a residue class)**.** *For $r \in \mathbb{Z}/q\mathbb{Z}$,*

$$\frac{1}{N}\#\{n \le N : n \equiv r \pmod q, \mu(n)^2 = 1\} \longrightarrow \delta_{q,r} := \sum_{\substack{d\ge 1 \\ (q,d^2)\mid r}} \frac{\mu(d)}{\mathrm{lcm}(q,d^2)}. \tag{9}$$

Proof. Use $\mu(n)^2 = \sum_{d^2|n} \mu(d)$. The simultaneous congruences $n \equiv r \pmod q$ and $d^2 \mid n$ are soluble exactly when $(q, d^2) \mid r$, and then have density $1/\operatorname{lcm}(q, d^2)$. Truncating at $d \leq \sqrt{N}$ gives an $O(\sqrt{N})$ counting error; the tail is $O(N/\sqrt{N})$. Absolute convergence of the displayed series completes the limit. □

Theorem 3.4 (One-site transducer correlation)**.** *Let a universally safe transducer satisfy* (8)*. Then its explicit Mobius path obeys*

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n\leq N} \mu(n)\ell(v_n) = L(T) := \frac{1}{2} \sum_{r \bmod q} \delta_{q,r}\big(g_r(1) - g_r(-1)\big). \tag{10}$$

Consequently,

$$|L(T)| \leq \liminf_{N\to\infty} \frac{K_N(G,\ell)}{N} \leq \limsup_{N\to\infty} \frac{K_N(G,\ell)}{N} \leq \frac{6}{\pi^2}\|\ell\|_\infty. \tag{11}$$

Proof. For each residue r, let $Q_r(N)$ be the squarefree count in (9) and put

$$M_r(N) = \sum_{\substack{n\leq N \\ n\equiv r \pmod q}} \mu(n).$$

Davenport's fixed-frequency estimate, followed by finite Fourier inversion, gives $M_r(N) = o(N)$. Hence the positive and negative counts in that residue are $(Q_r \pm M_r)/2$. Since zero Mobius values contribute zero to the correlation, the residue-r contribution divided by N is

$$\frac{Q_r}{2N}\big(g_r(1) - g_r(-1)\big) + \frac{M_r}{2N}\big(g_r(1) + g_r(-1)\big).$$

Apply the lemma and sum over the finitely many residues. The explicit path is admissible by (7), so its absolute score is bounded by the capacity; the upper bound is (6). □

Proposition 3.5 (Finite bounded-resource classification)**.** *Fix G, q, and a memory budget $m = |S|$. The set of all transducer tables satisfying universal safety and the one-site condition is finite and can be exhaustively enumerated. If*

$$\Gamma_{q,m}(G,\ell) = \max_T |L(T)|,$$

where the maximum is over that finite set, then

$$\Gamma_{q,m}(G,\ell) \leq \liminf_{N\to\infty} K_N(G,\ell)/N.$$

Proof. There are finitely many choices for the transition and output entries of the tables $S \times (\mathbb{Z}/q\mathbb{Z}) \times \mathcal{A}$, and finitely many initial states. Conditions (7) and (8) are finite Boolean tests. The inequality follows by maximizing the individual lower bounds in (11). □

Remark 3.6 (What the finite classification does not say)*.* The proposition is not a classification of all mixing subshifts: q and the memory budget are fixed before enumeration. If the observed label depends on s_n, the residue reduction is invalid and higher-order Mobius correlations enter. We do not replace those missing correlations by pair data or by a finite numerical fit.

4 Three frozen certificates

4.1 The RH-366 four-state label rule

Use the two-state universal safety completion supplied in the artifact. It has the same one-site labels as the RH-366 rule. Set

$$e_n = +1 \iff \mu(n) = 1 \text{ and } n \bmod 4 \in \{1, 2\},$$

and output a graph representative with that label. The state-1 rows at phases 2 and 3 are completion rows; they are not an additional arithmetic observable. The selected residues shift by two to $3, 0 \pmod 4$, so two selected plus signs can never be two places apart. The completed state table is universally safe and its observed factor has $g_1(1) - g_1(-1) = g_2(1) - g_2(-1) = 2$, with the other differences zero. Thus

$$L(T) = \delta_{4,1} + \delta_{4,2} = \frac{4}{\pi^2}.$$

This recovers the RH-366 lower certificate, not its unresolved capacity limit.

4.2 The RH-368 three-cell factor

For (2), at odd phases output vertex 0 when $\mu(n) = 1$ and vertex 1 otherwise; at even phases output vertex 2. Every output edge is legal. Since $\ell = (1, -1, -1)$, this has $g_1(1) - g_1(-1) = 2$ and gives

$$L(T) = \delta_{2,1} = \frac{4}{\pi^2}.$$

RH-368 proves the separate parity-factor capacity limit; here it is used as an independent frozen graph audit of the general certificate contract.

4.3 A new period-three safe switch

On the RH-366 graph, use the anchor state $0 = (--)$. At a phase $n \equiv 1 \pmod 3$, choose the loop

$$0 \to 2 \to 1 \to 0$$

when $\mu(n) = 1$, and choose

$$0 \to 0 \to 0 \to 0$$

otherwise. The two paths have equal length and their observable labels $(-1, 1, -1, -1)$ versus $(-1, -1, -1, -1)$ differ only at the first output position. The finite state table also defines safe outputs from every unreachable state, so the universal check is literal rather than trajectory-dependent. Therefore

$$g_1(1) - g_1(-1) = 2, \qquad L(T) = \delta_{3,1} = \frac{9}{4\pi^2}.$$

This is a new bounded certificate, not a claim about the RH-366 optimizer's limit.

5 Executable audit

The artifact uses a linear integer Mobius sieve, exact max-plus recurrences, literal universal table checks, and a small exhaustive table enumeration. It checks $N \leq 128$ for all three transducer witnesses, an endpoint $N = 2^{16}$, and all $3^6 = 729$ one-state output tables on the RH-368 graph at $q = 2$. The source manifest locks nine files and five source commits. These are finite reproduction and provenance checks. The only asymptotic steps are the squarefree-density lemma and the cited Davenport estimate used in the proof.

Instance	graph vertices	clock	memory	certified limit
RH-366 rule	4	4	2	$4/\pi^2$
RH-368 factor	3	2	1	$4/\pi^2$
RH-366 q=3 switch	4	3	2	$9/(4\pi^2)$

Table 1: Finite safe transducer certificates. The constants are exact one-site limits; they are lower bounds for the corresponding open capacities.

6 Route verdict and Gate ledger

Route A is `GO` narrowly: the max-plus theorem, the finite safety classification, the one-site arithmetic limit, and the three audited certificates form an independent theorem package. Route B is `STOP_SCOPED`. The transducer is an offline arithmetic selector, so the first mismatch occurs before Gate A: no canonical intrinsic operator, determinant, signed von-Mangoldt trace, or completed-zeta divisor is present. The RH-366 capacity limit remains open; memory-dependent labels remain outside the unconditional theorem.

The five project Gates remain false/open. This paper does not construct a Hilbert–Polya operator, identify Riemann zeros, prove a prime-power trace, or prove the Riemann Hypothesis.

7 Conclusion

Finite constraint graphs admit a precise arithmetic certificate layer: exact finite optimization, a decidable bounded transducer contract, and an unconditional one-site Mobius limit. The bounded qualifier is the result's strength and its boundary. Increasing memory or allowing the observable to read that memory is a new higher-order arithmetic problem, not something a finite table or pair ledger can silently solve.

References

[1] RH research program. Mobius orthogonality, adaptive encoding, and parry covariance, 2026. Repository paper RH-366, frozen release.

[2] RH research program. Parity-factor mobius capacity limit, 2026. Repository paper RH-368, frozen release.